

1.

Native - Built specifically for one operating system (iOS or Android) using dedicated languages like Swift, Objective-C, Kotlin, or Java. Use for High-performance demands, heavy graphics, gaming, complex hardware integration, or strict security needs.

Hybrid Development - Built using web technologies (HTML5, CSS, and JavaScript) and wrapped in a lightweight native shell to access basic phone features.

Use for Simple content-driven apps, minimum viable products (MVPs), tight budgets, or teams with existing web development skills.

Cross-Platform Development - Built using unified frameworks (such as Flutter or React Native) that compile into near-native components or use a native bridge.

Use for Standard business apps, e-commerce platforms, or projects needing simultaneous multi-platform launch with limited funding.

2.

Flutter Ecosystem is the massive collection of external tools, extensions, and libraries built by Google and the developer community to support Flutter.

Flutter Ecosystem helps developers build modern mobile apps using a single codebase, hot reload, and custom widgets. It allows teams to create fast, multi-platform applications for iOS, Android, and other systems from one place.

3.

The Software Development Life Cycle (SDLC) is a step-by-step process used by teams to plan, build, test, and deploy software. It relies on core stages like planning, coding, and testing to keep projects organized.

It is important because it reduces project costs, catches errors early, and ensures the final application meets user and business goals.

4.

Flutter development practices and the Software Development Life Cycle (SDLC) work together by providing both the tools and the process needed to build high-quality mobile applications. The SDLC serves as a structured framework that guides a project from planning to maintenance, while Flutter provides the technology to efficiently develop applications for Android and iOS using a single codebase.

5.

Developers should understand concepts such as Native, Hybrid, Cross-platform development, the Flutter ecosystem, and the Software Development Life Cycle (SDLC) before starting an application project because these concepts help them make informed decisions and build better applications. Knowing the strengths and limitations of each development approach allows developers to choose the most suitable technology based on the project's requirements, budget, timeline, and target users.